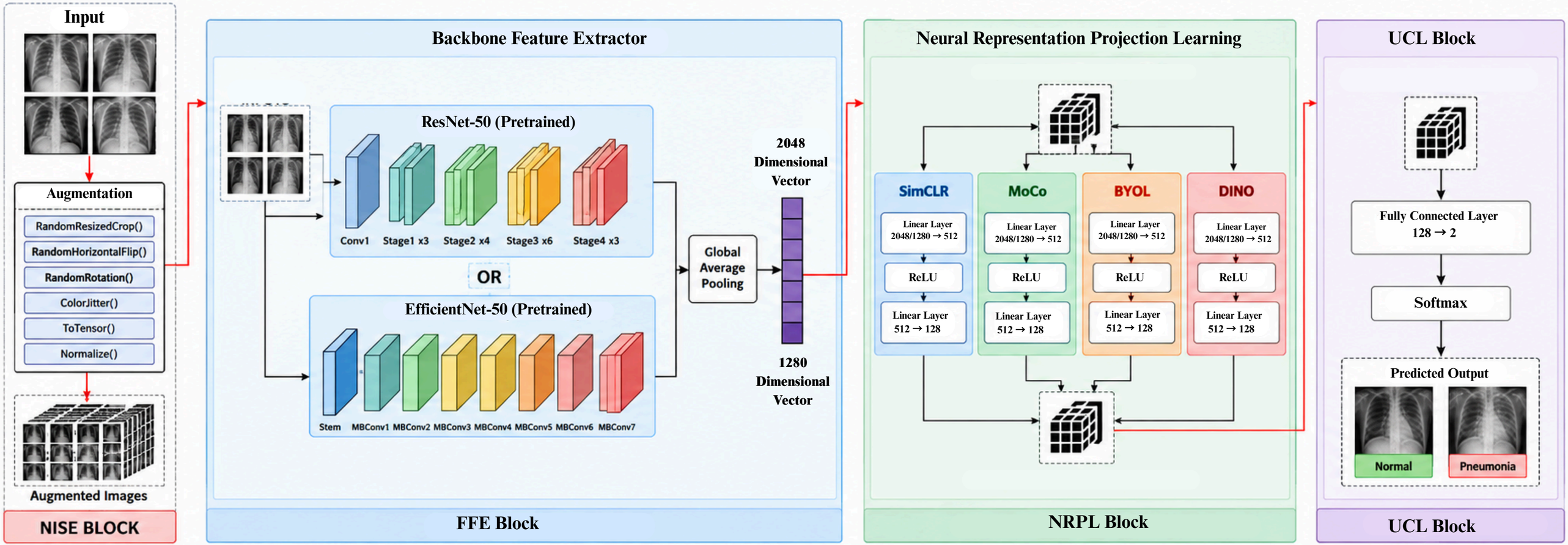

Input
Augmentation
RandomResizedCrop()
RandomHorizontalFlip()
RandomRotation()
ColorJitter()
ToTensor()
Normalize()
Augmented Images
NISE BLOCK
Backbone Feature Extractor
ResNet-50 (Pretrained)
Conv1
Stage1 x3
Stage2 x4
Stage3 x6
Stage4 x3
OR
EfficientNet-50 (Pretrained)
Stem
MBConv1
MBConv2
MBConv3
MBConv4
MBConv5
MBConv6
MBConv7
Global Average Pooling
2048 Dimensional Vector
1280 Dimensional Vector
FFE Block
Neural Representation Projection Learning
SimCLR
MoCo
BYOL
DINO
Linear Layer 2048/1280 → 512
ReLU
Linear Layer 512 → 128
NRPL Block
UCL Block
Fully Connected Layer 128 → 2
Softmax
Predicted Output
Normal
Pneumonia
UCL Block